Critical Review of Taxation Perspectives from Large Language Models: A Case Study on Additional Tax Penalties

Graduate Research Mini Paper

Student: Dai Phuong Ngo
CSCI E-222 Foundations of Large Language Models
Professor: Dmitry V. Kurochkin, PhD
Harvard Extension School, Harvard University
May 2026

Choi et al., Taxation Perspectives from Large Language Models: A Case Study on Additional Tax Penalties (EACL 2026)

https://aclanthology.org/2026.eacl-long.218/

For this Graduate Research mini-paper for CSCI E-222 Large Language Models, I chose Choi et al.'s *Taxation Perspectives from Large Language Models: A Case Study on Additional Tax Penalties*, published at EACL 2026. I ended up picking this paper because it felt to me more relevant to my domain and interests where I have been working or than other recent LLM papers I looked at. Since I am interested in the kind of work done in a Big 4 setting, especially around corporate tax filing, reconciliation and advisory, I wanted something that was not just about general chatbot performance. What attracted me to this paper is that it concentrates on a very practical and realistic question: can large language models actually handle tax reasoning and calculation in a way that resembles professional analysis or are they mainly just good at making their answers sound convincing without high realiability?

My paper mentions a benchmark called PLAT, built from 100 Korean court precedents with involvement of additional tax penalties. From its cases, the authors create 300 total examples including 100 binary-choice questions, 100 multiple-choice questions and 100 essay questions. I used to believe that this choice was strong enough for testings and applications because it makes the assessment more relevabt realistic than a standard benchmark with short and simple prompts. Tax analysis is rarely just about choosing one answer off a list. In real settings, the answer usually depends on context, facts, interpretation and how professionally tax experts can explain the reasoning behind the conclusion. That is what my paper is trying to assess.

One thing I was attracted is that the paper is not only asking whether an LLM knows tax law. It is asking whether the LLM can reason through a tax problem in a structured way. The IRAC framework, standing for Issue, Rule, Application and Conclusion, is used by the authors. I consider this part outstanding because IRAC is very close to how legal and tax analysis is actually executed in practice. A professional does not just state a rule and stop there. Tax experts must identify the issue, find the right authority, apply that authority to the specific facts and then defend a conclusion. This paper illustrates that this is where LLMs still struggle the most, especially in the Application and Conclusion stages.

In terms of method, the paper is pretty comprehensive. The authors asses ten different models, including both open and commercial systems. These include Qwen3-32B and EXAONE3.5-32B, along with GPT-4o, GPT-4.1, o1, o3, o3-mini, Claude 3.7 Sonnet and Gemini 2.5 Flash and Pro. They also test retrieval-augmented generation, also known as RAG, using BM25 retrieval over a set of precedents and statutory articles. Beside that, the essay responses are scored using ten rubrics built by tax experts and aligned with IRAC. I highly value that the evaluation was not reduced to one simple number, because that would not really capture what makes tax reasoning difficult.

At first, some of the results look quite strong. On the PLAT multiple-choice task, o3 and Gemini 2.5 Pro gain the highest F1 score at 0.79, which might sound impressive to me, and when comparing to older LLM performance in specialized domains, it might be so. But what makes the paper more interesting to me is that the authors do not stop at the headline result. They split the cases into easier and harder ones, and that bring more insights. The models perform better on easy cases, where the rule points more directly to an answer. However, they perform worse on hard cases, where the model must deal with ambiguity, taxpayer-specific facts or competing or conflicting principles. To me, that is one of the most important findings in the paper. A model can look strong on average and still be unreliable when the problem becomes more realistic and complex.

That point also matters a lot in a professional context. In a place like consulting firms, the easy cases are not usually the biggest problem. The higher-value work is often in the gray areas, where the facts are messy and the answer is not obvious while tax rules and regulations might changes over time that consulting firms have to publish tax facts every year. So even though the paper shows that LLMs are getting stronger, it also demonstrates why benchmark scores should be interpreted carefully. A high score does not necessarily mean a model is dependable in the cases that matter most.

Another oart of the paper I found useful was the discussion of retrieval-augmented generation. In many business conversations, RAG is considered and treated almost like a cure-all. The thinking is that once the model has access to the right documents, its answers will become trustworthy. This paper suggests that things are not that straighforward and simple. Some models improve only slightly with retrieval, while others do not improve at all or even perform worse. The paper provides two possible explanations for these behaviors. One is that the retrieval system may not be finding the most relevant legal materials. The other, which I think is even more important, is that the real weakness is not just in finding the rule. The bigger weakness is in applying the rule correctly to the facts and then reaching a sound conclusion. I think that is a valuable result to consulting context as it forces back against the idea that better document access naturally or automatically solves reasoning problems right away.

I also used to think that the PLAT-MCR experiment was interesting. In that setting, the multiple-choice answers include rationales along with the conclusions. When the models are given those more structured options, accuracy increases quite a bit but not so obviously. To me, that suggests the models are often better at recognizing a plausible line of reasoning than generating one from scratch. That difference matters. It implies that, at least for now, LLMs may be more useful as support tools than as fully independent decision-makers. In a tax setting specifically, that could mean using them to compare possible interpretations, organize issues, suggest numerical figures and calculations or help draft an initial memo, while leaving the final judgment to a human professional.

The essay evaluation is another strong part of the paper. The essay benchmark is scored with ten rubrics designed by tax professionals. And the paper also compares LLM judging with human expert grading. I liked this because it makes the evaluation feel more grounded in actual domain expectations. At the same time, the authors expressed honesty about the limits. They note that the LLM judge is somewhat more generous than human graders and is not equally reliable across all rubric categories, especially when legal authority must be checked carefully. I think that honesty improves the paper. I think that this one is cautious about what the results really mean.

Regarding strengths, I think the biggest one is realism. It focuses on a domain where facts matter, rules can conflict and compete, and good reasoning matters more than surface fluency. I also think the IRAC framing makes the analysis much more useful. Instead of treating every error the same way, the paper tries to identify where the reasoning breaks down. That makes the results more informative, especially for people who pay more attention to real-world applications.

In my viewpoint, the main limitation is scope. The benchmark is based on Korean additional-tax-penalty cases. Therefore, its findings should not be generalized too broadly. That does not make the paper weak, but it does mean the results need to be interpreted carefully. I would not take this paper as proof of how LLMs will perform in every tax setting, especially not in U.S. or Canadian corporate tax or asset management. Different jurisdictions, legal systems and fact patterns could produce different results. Still, I do think the paper captures something important that probably carries over across domains: LLMs tend to perform better when the task looks like rule matching and worse when the task requires deeper judgment.

This paper also connects well with several themes from the course. It touches on domain specialization, hallucination, retrieval-augmented generation, reasoning models and LLM-based evaluation. More broadly, it shows why we need to be careful when interpreting model performance. A model can be very fluent and still fail in subtle but important ways. That is true in high-stakes domains, where making it sound correct is not the same thing as actually making it correct.

My main takeaway from the paper is that LLMs seem promising as decision-support tools in tax, but not as replacements for professional judgment. They can probably help with issue spotting, organizing authority, and drafting a first-pass analysis. But when a case depends on facts with nuances, judgment calls or careful legal-rule-based application, the models still seem unreliable. To me, that is the most practical lesson in this paper. This paper is not saying LLMs are useless or not too useful in tax. It is saying their strengths and limits need to be understood clearly before people treat them as dependable experts.

Overall, I think this is a strong paper for the mini-paper assignment because it is recent, published in a reputable venue, clearly connected to course themes and directly relevant to a real tax consulting domain. More importantly, it does not just demonstrate where LLMs perform well. It also demonstrates where they fall short and that is what makes it worth writing about and reading. The paper's biggest contribution, in my opinion, is not that it proves LLMs can do tax reasoning well. It is that it illustrates how much harder real tax reasoning is than simply retrieving a rule or producing a fluent answer. That makes it both academically interesting and professionally relevant.